Shubham Jolapara

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Summary

Results-driven Robotics Engineer with 2+ years of experience building and integrating real-world autonomous robots — spanning perception, localization, control, and embedded execution. Proficient in ROS/ROS2, C++, Python, and computer vision (OpenCV, PyTorch, YOLOv5). Hands-on experience with PID control, Kalman-filter-based odometry, visual-inertial localization, and edge deployment on Linux/Ubuntu.

Education

Gujarat Technological University — Ahmedabad, India
B.E. in Computer Science and Engineering ***(CGPA: 7.99/10)*** — July 2020 – June 2024

Work Experience

Indrones Solutions Pvt. Ltd. — Navi Mumbai, India
R&D Engineer — Aug 2024 – Present

Insynq Ground Control Station

- Extended **QGroundControl** (C++, QML, MAVLink) to build the cross-platform Insynq GCS, leveraging Qt's **multi-threaded architecture** for real-time telemetry; added dual-joystick instructor/trainee control, KML/KMZ mission import, and selectable CRS coordinate indicators.
- **Firmware integrity verification** during flashing, including bootloader-based board detection and checksum validation (APJ SHA-256, BIN, Intel HEX); enhanced bootloader protocol to log boot observations to `bootlog.txt` on the SD card.
- Engineered **MAVLink-based gimbal POI visualisation** by periodically requesting CAMERA_FOV_STATUS messages and rendering live lat/lon overlays on the Fly View map.

DOOAF – Direction of Own Artillery Fire

- Developed a geospatial targeting system to estimate ground targets from a drone's **gimbal pointing direction**, addressing an Indian Army artillery observation requirement.
- Projected the **camera optical axis** to the ground, intersecting with **ArduPilot terrain DAT** elevation data; achieved target accuracy of **up to 50 m**.

ArduPilot Firmware Contributions

- Exposed firmware verification results (CRC, signature validation, board compatibility) via the bootloader protocol; automated signing and SHA-256 checksum appending in the **Waf build system** via configurable build flags.
- Added support for new hardware platforms including bootloader updates and hardware definition files for **STM32-based autopilots**.
- Developed internal **BMS driver (BQ34100)** for battery monitoring; upstreamed **Benewake TFS20L** LiDAR rangefinder driver as an open-source contribution to ArduPilot.

ArduPilot ROS2 Simulation & Autonomy

- Built an **ArduPilot + ROS2** simulation stack in **Gazebo** with multi-camera support (gimbal and nadir); developed ROS2–Gazebo bridge pipelines for camera, TF, and ground-distance sensor topics.
- Created a **ROS2 RTSP streaming node** (GStreamer, H.264, 30 fps) bridging simulation camera topics to network streams for live remote monitoring.

GPS-Denied Navigation Research

- Constructed a handheld **visual-inertial sensing platform** (Raspberry Pi 4, Pi Camera, VectorNav IMU) for navigation in GPS-denied environments.
- Deployed **OpenVINS** on-device for visual-inertial SLAM (VI-SLAM) and trajectory estimation; performed camera-IMU calibration and hardware-level sensor synchronisation.
- Evaluated pose estimation accuracy through indoor experiments, diagnosing sensor drift and timing failures in the field.
- Built a non-GPS **VIO evaluation pipeline** coupling the prebuilt **AirSimNH** simulator binary with **ArduPilot SITL Copter**; authored a **ROS2 bridge** relaying simulated downward-facing camera and IMU streams to OpenVINS' `ov_msckf` estimator.

- Benchmarked OpenVINS-estimated trajectories against **ArduPilot ground-truth pose** across 3D autonomous flight profiles, reporting **absolute trajectory error (ATE)** and drift.

IIT Gandhinagar
Gandhinagar, India
Research Intern
April 2024 – July 2024

- Applied **Physics-Informed Neural Networks (PINNs)** to solve PDEs (Burgers' equation benchmarks), extending the framework to predict **temperature distribution during laser metal carving** using physics constraints alone, eliminating reliance on experimental data.
- Engineered a two-stage **training pipeline** incorporating **transfer learning** to accelerate model convergence and improve prediction accuracy.

Competitions & Projects

ABU Robocon 2023
Phnom Penh, Cambodia
Software Team Member, GTU Robotics Club
Aug 2023

- Represented **Team India** at ABU Robocon 2023 after winning the national round.
- Applied **Extended Kalman Filter (EKF)** for accurate robot pose estimation and localisation.

DD Robocon 2023
IIT Delhi, India
Software Team Member, GTU Robotics Club
July 2022 – June 2023

- Built software for a **semi-autonomous robot** with tri-wheel holonomic drive and ring-picking/shooting mechanism.
- Implemented **odometry** using encoders to accurately track and **localize** the robot's position, using sensor data to estimate changes in position over time.
- Diagnosed failure modes in motor control and state estimation under competition pressure; iterated rapidly on controller tuning to meet task requirements.

E-Yantra 2023 – HolA Bot
IIT Bombay, India
IIT Bombay Competition
Aug 2022 – March 2023

- Simulated a **3-wheeled omnidirectional base** in **Gazebo** with a closed-loop "Go to Pose" controller for accurate pose control.
- Achieved **trajectory execution**: Function Mode traces parametric paths $x(t), y(t), \theta(t)$; Image Mode follows contours extracted from a 640$\times$360 image.
- Used overhead camera feedback for closed-loop **path correction** and continuous pose refinement.

DD Robocon 2022
IIT Delhi, India
Technical Team Member, GTU Robotics Club
Aug 2021 – July 2022

- Implemented **PID controller** for real-time holonomic drive control using IMU and LiDAR sensors; tuned parameters under live competition conditions.
- Built a complete **perception-to-action pipeline**: Intel RealSense depth camera with **YOLOv5** target detection driving an autonomous turret positioning and shooting actuator; stack: PyTorch, OpenCV, Arduino.

Skills

Programming Languages: C, C++, Python
Robotics & Frameworks: ROS, ROS2, Gazebo, OpenVINS, ArduPilot, MAVLink, Qt/QML
Navigation & Control: Visual-Inertial Odometry (VIO), Kalman Filter, Odometry, PID Control, Holonomic Kinematics
Perception / Vision: OpenCV, YOLOv5, PyTorch, Intel RealSense, NumPy, pandas
Embedded & Hardware: Raspberry Pi 4, STM32, Arduino, ESP32, BLDC control, UART/SPI/I2C, IMU, LiDAR, depth cameras
Software & Tools: Git, Linux (Ubuntu), STM32CubeIDE, SSH

Achievements

- Represented **Team India** at **ABU Robocon 2023**; achieved **6th Rank** globally.
- Achieved **AIR 1** out of 43 teams in **DD Robocon 2023**.
- Achieved **AIR 10** out of 45 teams in **DD Robocon 2022**; won **Visvesvaraya Best Design Award**.
- Cleared **E-Yantra 2023** Stage 3 with a perfect score of **100/100**.